

Plotting, Ordering and Comparing Real Numbers

Circuit 7A.02.06

Name _____

Date _____

Period _____

Work the four parts in order, on your own. Everything here is from Lessons 1 through 5. This is the last practice before the unit assessment.

PART 1. Comparing

Use $<$, $>$ or $=$. One of the six is an equality.

1.	-5π		$-\sqrt{225}$
2.	$\sqrt[3]{125}$		$\frac{\sqrt{100}}{2}$
3.	$\sqrt{45}$		$\pi + 4$
4.	-7		$-\sqrt[3]{49}$
5.	$\frac{29}{3}$		$\sqrt{80}$
6.	$\frac{\pi}{3}$		$\sqrt{22} - 3$

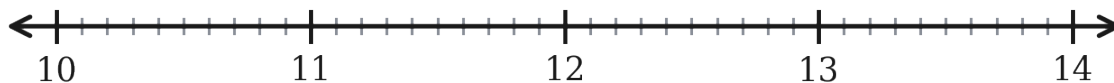
PART 2. Sorting and plotting

A number is irrational only if it cannot be written as a ratio of two integers.

7. Sort the five expressions, then plot and label each one on the number line.

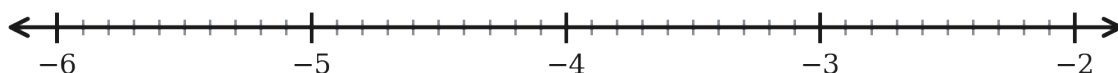
$$10.2 \quad 11\frac{1}{3} \quad \sqrt[3]{64} + 8 \quad -\pi + 16 \quad \sqrt{190}$$

Only rational numbers	Includes an irrational number



8. Plot and label each expression on the number line.

$$-\frac{60}{11} \quad \sqrt[3]{-125} \quad -\sqrt{20} \quad -\frac{7}{2} \quad -\pi \quad -\sqrt{8}$$

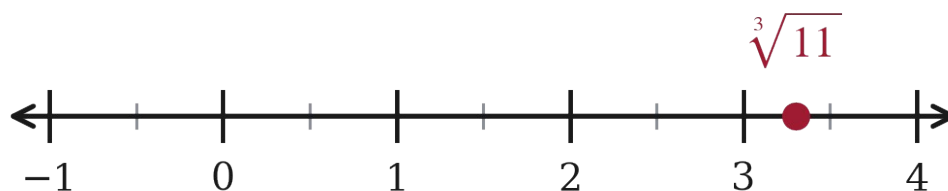


PART 3. Error analysis

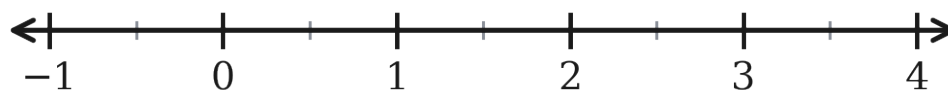
Read the work, find the error, then do it correctly.

9. Sara approximated $\sqrt[3]{11}$ like this:

“11 is between 9 and 16, which are both perfect squares. So $\sqrt[3]{11}$ lies between 3 and 4, and it is closer to 3 because 11 is closer to 9 than to 16.”



- What is Sara's error?
- Redo it correctly, then plot the right value.



PART 4. Ordering and plotting

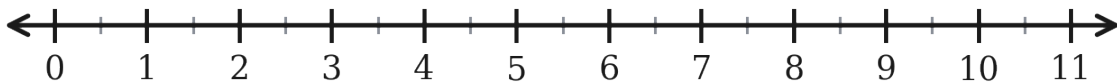
10. Order the numbers from least to greatest, then plot and label each one.

$$\sqrt[3]{8} \quad \pi \quad \sqrt[3]{100} \quad \sqrt{48} \quad \frac{19}{2} \quad \sqrt{121}$$



11. Order the numbers from least to greatest, then plot and label each one.

$$\pi - 3 \quad \sqrt[3]{49} \quad \sqrt[3]{125} \quad 7\frac{1}{2} \quad \sqrt{80} \quad \sqrt{115}$$



STRETCH. Beyond the range the test uses. Same method, bigger cubes.

12. Find the value of $\sqrt[3]{216}$ and of $\sqrt[3]{-343}$.